Lab 2 – DNS, IP, and Wireshark Analysis

Question 1

Protocol Data Units (PDUs)

- name for layer 4 TCP PDU is Segment / TCP
- name for layer 4 UDP PDU is Datagram / UDP
- name for layer 3 PDU is Packet/IP
- name for layer 2 PDU is Frame/wifi/Ethernet
- IP Address (IPv4): 10.10.99.19
- MAC Address: 40-1A-58-E8-B1-D2

this is from the Wi-Fi adapter, the only active interface.

Question 3

- DNS Servers: 85.197.128.3 and 85.197.129.3
- Used to translate domain names.
- Computer can contact the correct server.
- DNS resolves human-readable names to IP addresses.

Question 4

- Server: c.resolvers.bikab.com
- Address: 85.197.128.3

IP addresses for www.mit.edu

- 2a02:26f0:a00:486::255e
- 2a02:26f0:a00:4b5::255e
- 23.40.225.75

Is www.mit.edu an alias?

- Yes it is an alias.

{Correct — www.mit.edu is a CNAME pointing to Akamai.}

Question 5

Authoritative DNS servers for mit.edu:

- usw2.akam.net
- ns1-173.akam.net
- asia1.akam.net
- use5.akam.net
- asia2.akam.net
- ns1-37.akam.net
- use2.akam.net
- eur5.akam.net

Their IP addresses:

- usw2.akam.net → 184.26.161.64
- ns1-173.akam.net → 2600:1401:2::ad, 193.108.91.173
- asia1.akam.net → 95.100.175.64
- use5.akam.net → 2600:1403:a::40, 2.16.40.64
- asia2.akam.net → 95.101.36.64
- ns1-37.akam.net → 2600:1401:2::25, 193.108.91.37
- use2.akam.net → 96.7.49.64
- eur5.akam.net → 23.74.25.64

Question 6

- Server: dns.google
- Address: 8.8.8.8

IPv6 for www.hkr.se:

- No, there are no IPv6 addresses available.
- HKR only provides IPv4 A records.

Question 7

- No, long name. It is a reverse DNS (PTR) record. Used mainly by ISPs, mail servers, network administrators, security systems.
- end users never type PTR names, so no inconvenience.

Question 8

- hkr-se.mail.protection.outlook.com
- Mail is delivered to Microsoft's Outlook / Exchange Online Protection mail server.
- HKR uses Microsoft 365 mail filtering.

Question 9

IPv6 addresses for www.google.com:

- 2001:4860:4826:7700::
- 2001:4860:4827:7700::
- 2001:4860:4828:7700::
- 2001:4860:4829:7700::
- 2001:4860:482a:7700::
- 2001:4860:482b:7700::
- 2001:4860:482c:7700::
- 2001:4860:482d:7700::

DNS Load Balancing Explanation:

- DNS based load balancing works by returning multiple IP addresses for the same domain name.
- Different DNS resolvers reorder those IPs so users connect to different servers.
- This spreads traffic across global infrastructure and prevents overload.
- Improves performance and reliability.

Question 10

- There are a lot of domains, 120 entries, most are repeated PTR records.
- This means output redirection operator directs to open and read the file.
- Windows DNS cache often contains many PTR entries.

Question 11

- **Packet numbers:** 1594 (Query) and 1600 (Response)
- **Protocol:** UDP
- **Reason:** DNS typically uses UDP for queries and responses because it is faster and involves less overhead than TCP. TCP is only utilized if the response size exceeds the 512-byte limit of a standard DNS UDP packet (or if performing zone transfers).

Question 12

Source and destination ports:

- **Source and Destination Ports:**
- **Query (1594):** * Source Port: **53283**
    - Destination Port: **53**
- **Response (1600):** * Source Port: **53**
    - Destination Port: **53283**

Question 13

- **Sent to: 85.197.128.3**
- **Reason:** This is the destination IP address of the query packet. In this network environment, this IP represents the DNS resolver responsible for handling the name resolution request.

Question 14

Type: A (Host Address) Contains answers? No A DNS query packet only contains the "Question" section, which includes the Name (e.g., the URL being looked up), Type, and Class. The "Answer" section is empty in the query.

Question 15

- **Number of answers: 1 Each answer contains:** * **Name:** (The domain name requested)
- **Type:** A (IPv4 Address)
- **Class:** IN (Internet)
- **Time to Live (TTL):** (The duration the record is cached)
- **Data Length:** 4
- **Address:** (The actual IPv4 address) **IP in ASCII? No.** The IP address is transmitted in binary format. **Bytes for IPv4 address: 4 bytes.**

Question 16

- **DNS response size: 81 bytes**